

Starship Takeoff



This works in a locally installed editor, such as Thonny or the Raspberry Pi online editor:
• editor.raspberrypi.org

1

You are a starship captain. You have crashed your ship on a strange planet and must take off again quickly in the alien ship you have captured. The ship's computer tells you the gravity on the planet. You must guess the force required for a successful take off. If you guess too low, the ship will not lift off the ground. If you guess too high, the ship's fail-safe mechanism comes into operation to prevent it being burnt up. If you are still on the planet after ten tries, the aliens will capture you.

2

For the very first version, make a very easy guessing game. This is very easy because the number to guess is always the same!

- Set the secret variable to 7 (or some other number).
- Let the player guess.
- Check if the player has guessed correctly.

```
secret = 7

guess = int(input("> "))
if guess == secret:
    print("Correct!")
else:
    print("Wrong!")
```

Note: `input("> ")` lets the user enter a guess but what is `int` there for?

In a computer program there are different types of variable and it is important to make sure that they do not get mixed up.

- The `input` function returns a **string** of characters so the user can enter "123" and could also enter "word".
- An **integer** is a whole number and in the code above `secret` is an integer variable.
- To test `guess` with `secret` they need to both be the same type of variable. The `int` function changes a string into an integer.
- Later on there will be another variable type called a **boolean** that is always true or false. Can you spot it?

3

Lets make the game a bit more interesting:

- Tell the player if the guess is too high or too low.
- Give the player 10 chances to get it right.

```
secret = 7

for attempt in range(10):
    guess = int(input("> "))
    if guess < secret:
        print("Too low!")
    elif guess > secret:
        print("Too high!")
    else:
        print("Correct!")
        break
```

The `for` loop repeats the code to ask for a guess 10 times but if the player guesses correctly then the `break` command stops the loop immediately.

4

So far this is just a number guessing game. Lets add the space theme and make the number to guess a little more complicated.

- Set the gravity of the planet the player is trapped on. The player knows what the gravity is.
- Then set the mass of the spaceship. The mass is unknown to the player.
- The `required_force` that the player needs to use is then the gravity times mass. This is the number to guess instead of the secret.

```
gravity = 7
mass = 5
required_force = gravity * mass
print("Gravity is", gravity)

for attempt in range(10):
    force = int(input("Enter launch force: "))
    if force < required_force:
        print("Too low!")
    elif force > required_force:
        print("Too high!")
    else:
        print("Well done! You took off!")
        break
```

What is mass?

Mass tells us how much stuff something is made of. A spaceship with more mass is harder to launch than one with less mass.

Gravity also matters. Stronger gravity pulls the spaceship down more strongly. The **force** is the push from the spaceship's engines and it is worked out as:

$$\text{force} = \text{gravity} \times \text{mass}$$

5

Finally, to make the game more interesting, the gravity and mass will be chosen randomly.

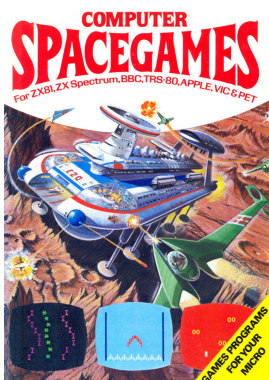
- The gravity and mass are chosen randomly using the `randrange` function from the `random` library.
- Also, add a message if the player gets caught by the aliens. To check this we need a new variable, `launched`, that is `False` when the spaceship is on the ground and `True` when the player managed to take off.

```
from random import randrange

gravity = randrange(1, 20)
mass = randrange(1, 40)
required_force = gravity * mass
launched = False
print("Gravity is", gravity)

for attempt in range(10):
    force = int(input("Enter launch force: "))
    if force < required_force:
        print("Too low!")
    elif force > required_force:
        print("Too high!")
    else:
        print("Well done! You took off!")
        launched = True
        break

if not launched:
    print("Bad luck! The aliens caught you!")
```



In the 1980s many young people learned to program by typing games from books like **Computer Spacegames**. This project recreates one of those games in modern Python.

The Computer Spacegames book, and various others, can still be found here:

<https://usborne.com/gb/books/computer-and-coding-books>